

- 24** Soap scum is a white residue that forms when soap reacts with minerals such as Ca^{2+} in hard water. Soap scum forms when the concentration of Ca^{2+} ions in water is greater than $10^{-5} \text{ mol dm}^{-3}$.

What is the minimum mass of sodium carbonate that should be added to 1 dm^3 of water to prevent soap scum from being formed?

$$[K_{\text{sp}} \text{ CaCO}_3 = 5.0 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}]$$

A 0.037 g

B 0.042 g

C 0.050 g

D 0.053 g